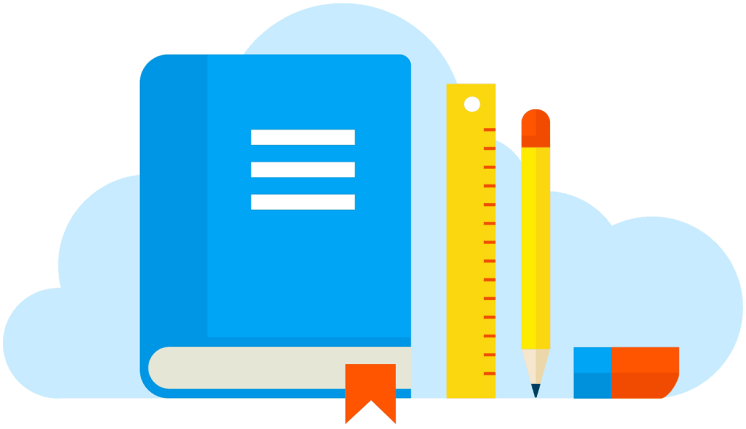


📘 NOTE 9 — NETWORK SECURITY FUNDAMENTALS

Date: 20/04/2025

1. Introduction to Network Security

Network Security is the process of protecting computers, networks, applications, and data from unauthorized access, attacks, and misuse.

The main purpose of network security is to protect:

- Data
- Network devices
- User accounts
- Applications
- Servers
- Communication

Example

In StudySphere, student login information, notes, assignments, and submissions should be protected from unauthorized users.

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2. Goals of Network Security

The main goals of network security are commonly explained using the **CIA Triad**.

1. Confidentiality

Only authorized users should be able to access information.

Example:
A student's personal profile should not be visible to unauthorized users.

2. Integrity

Data should not be changed or modified without permission.

Example:
A student's assignment grade should not be changed by another student.

3. Availability

Systems and data should be available when authorized users need them.

Example:
Students should be able to access StudySphere when submitting assignments.

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3. Common Network Security Threats

A **security threat** is anything that can potentially harm a network or its data.

Common threats include:

- Malware
- Viruses
- Phishing
- Password attacks
- Unauthorized access
- Denial-of-Service attacks
- Data theft
- Man-in-the-Middle attacks

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4. Malware

Malware means **Malicious Software**.

It is software designed to damage systems, steal information, or perform unauthorized activities.

Types of Malware

- Virus
- Worm
- Trojan
- Ransomware
- Spyware

Example

A user downloads an infected file and the malware damages files on the computer.

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5. Phishing

Phishing is a social engineering attack in which attackers try to trick users into providing sensitive information.

Attackers may use:

- Fake emails
- Fake websites
- Fake login pages
- Fraudulent messages

Example

A user receives a fake message asking them to enter their StudySphere password on a fake website.

Prevention

- Check website addresses carefully
- Do not open suspicious links
- Never share passwords
- Use multi-factor authentication when available

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6. Firewall

A **firewall** is a security system that monitors and controls network traffic based on security rules.

It can allow or block traffic.

Simple Working

Internet → Firewall → Network

The firewall checks incoming and outgoing traffic and decides whether it should be allowed.

Functions

- Blocks unauthorized traffic
- Controls network access
- Helps protect computers from attacks
- Monitors network connections

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7. Encryption

Encryption is the process of converting readable data into an unreadable form called **ciphertext**.

The original readable data is called **plaintext**.

Simple Flow

Plaintext → Encryption → Ciphertext

To recover the original data:

Ciphertext → Decryption → Plaintext

Example

A password or sensitive information can be protected while being transmitted over a secure connection.

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8. Authentication

Authentication is the process of verifying the identity of a user.

Common methods:

- Username and password
- OTP
- Biometrics
- Security keys
- Multi-factor authentication

Example in StudySphere

A student enters:

Email + Password → Authentication → Student Dashboard

Only an authenticated user should be allowed to access protected pages.

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9. Authorization

Authorization determines what an authenticated user is allowed to access or perform.

Example

In StudySphere:

Student:

- View notes
- Download materials
- Submit assignments
- View submission status

Admin:

- Manage students
- Add notes
- Create assignments
- View submissions

- Manage grades

Therefore, authentication answers:

“Who are you?”

Authorization answers:

“What are you allowed to do?”

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10. Antivirus

Antivirus software detects, prevents, and removes malicious software from computers.

Functions:

- Scans files
- Detects malware
- Removes threats
- Provides real-time protection
- Scans downloaded files

Example

Before opening a downloaded PDF or application, antivirus software can scan the file for malicious content.

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11. Access Control

Access control determines which users can access specific resources.

It is generally based on:

- User identity
- Role
- Permissions
- Security policies

Example

A normal student should not be allowed to open the admin dashboard.

Student → Student Dashboard

Admin → Admin Dashboard

This is called **role-based access control (RBAC)**.

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12. Digital Signature

A **Digital Signature** is a cryptographic technique used to verify the **authenticity and integrity** of digital information.

It helps answer:

- Who sent the data?
- Was the data modified?

Uses:

- Digital documents
- Online transactions
- Software distribution
- Secure communication

A digital signature provides stronger assurance that a message came from the claimed sender and was not altered.

13. Digital Certificate

A **Digital Certificate** is an electronic document used to associate an identity with a public key.

It is issued by a trusted **Certificate Authority (CA)**.

Commonly used for:

- Website authentication
- HTTPS connections
- Secure communication

Example

When a browser connects to an HTTPS website, the website uses a digital certificate to help prove its identity.

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14. Digital Signature vs Digital Certificate

Digital Signature

- Verifies data integrity
- Helps verify the sender
- Uses cryptographic techniques
- Applied to digital data/documents

Digital Certificate

- Helps verify the identity of a website or entity
- Contains identity information and a public key
- Issued by a Certificate Authority

- Commonly used with HTTPS

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15. Man-in-the-Middle Attack

A **Man-in-the-Middle (MITM) attack** occurs when an attacker secretly intercepts communication between two parties.

Simple Example

Student → Attacker → StudySphere Server

The attacker attempts to intercept the communication between the student and server.

Prevention

- Use HTTPS
- Use secure networks
- Avoid suspicious Wi-Fi networks
- Keep systems updated

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16. Denial-of-Service Attack

A **Denial-of-Service (DoS) attack** attempts to make a service unavailable by overwhelming it with requests or traffic.

Example

An attacker sends a large number of requests to a server, making it difficult for legitimate users to access the website.

Prevention

- Firewalls
- Traffic filtering
- Rate limiting
- Monitoring
- DDoS protection services

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17. Network Security in StudySphere

Security is important in StudySphere because the system contains student and academic information.

Important security measures:

1. Authentication
Students and administrators must log in.

2. Authorization
Different roles have different permissions.

3. Password Protection
Passwords should be securely stored using appropriate hashing techniques.

4. HTTPS
Protects communication between browser and server.

5. Input Validation
Helps prevent invalid or malicious input.

6. Access Control
Students should not access admin-only functions.

7. Database Security
Database access should be protected from unauthorized users.

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18. Basic Security Practices

Users should follow these practices:

- Use strong passwords
- Do not share passwords
- Keep software updated
- Avoid suspicious links
- Use antivirus software
- Use secure networks
- Enable multi-factor authentication where available
- Back up important data
- Log out from shared computers
- Use HTTPS websites for sensitive activities

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19. Simple StudySphere Security Flow

Student

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Login

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Authentication

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Authorization / Role Check

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Student Dashboard

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Secure Server Communication

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Database

This ensures that users can access only the features permitted for their role.

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20. Quick Revision

Remember these important terms:

Network Security → Protects networks and data

CIA Triad → Confidentiality, Integrity, Availability

Firewall → Controls network traffic

Encryption → Converts plaintext into ciphertext

Authentication → Verifies user identity

Authorization → Determines user permissions

Antivirus → Detects and removes malware

Access Control → Controls resource access

Digital Signature → Helps verify sender and data integrity

Digital Certificate → Associates identity with a public key

Phishing → Tricks users into revealing information